Wednesday, June 17, 2026 6:30 AM

BOOSTING

Agenda ⇒

- → Recap
- → Boosting
- → Training & testing
- → Pseudo Residuals
- → Sklearn implementation

Recap → Ensemble ⇒ DT → multiple DTrees

pred^ → Become more stronger

R.S + C.S

Bagging → Bootstrap + Aggregation

Random Forest

RF = Base learners + Randomisation + Aggregation

DTrees

Overfit

High variance

low Bias

R.S + C.S

↓

to predict

Estimate value of House ⇒

House price = 1 CR

5 people

(A) → 90 lac
(B) → 110 lacs
(C) → 95 lacs
(D) → 105
(E) → 100

avg = 1 CR

(x) max_depth = 2 ⇒

(B) Tree 1 → 80
2 → 82
3 → 81
4 → 84
5 → 80

Simple

⇒ 81.2

→ High Bias

all of them are wrong

max_depth = None ✗

Tree 1 → 90
2 → 110
3 → 95
4 → 105
5 → 100

↳ Overpredicting
↳ underpredicting

100 = 1 CR

100 doctors ⇒

Scenario 1

18%
12%
18%

Rule set

Scenario 2 →

D6 → 10%
2 → 14%
3 → 18%

≃ actual

(4) Single Tree → low Bias · High var

(5) Forest → avg → Low Bias
Low Variance

(6) Boosting → ensemble type

(1) Boosting → ensemble type

lot of trees → diff. models

Base learners → DTrees

Boosting Vs Bagging

Boosting

Base learners → low var., high bias

→ underfitting

Bagging

Base learners → OVERFITTED

(2) Example of underfit model ⇒

(1) max_depth = 1

(2) Avg model / Mean model

simple

↓

underfit

* Boosting → Base learners → low var, high bias

* Boosting → Base learners → High Bias (underfit)
  → DTrees = low depth

(*) Bagging → introduced Randomisation
  → They do not have High variance
  → (ND)

→ "Additive combining"
  * Reduce Bias

⇒

Decision Tree → underfit

Train → max_depth = 1

ND good performance

Which of the following model are underfit models ?

1 user has participated

✓	A	Decision Stump (Decision Tree with depth = 1)	100%
	B	Decision Tree with depth = 20	0%
	C	Decision Tree with depth = 10	0%
	D	None of the above.	0%

End Quiz Now

Quiz time!

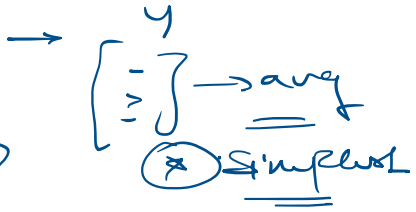
Time Left: 8s

What model will have high bias low variance ? (Hint: think of simplest model)

40 users have participated

✓	A	Mean model (which predict mean value everytime)	63%
	B	Random model (model which predicts random value everytime)	20%
	C	A DT fit perfectly on training data	17%

End Quiz Now



Quiz time!

Time Left: 7s

What will be the training error for high bias model ?

✓	A	High error	78%
	B	No error	2%
	C	Low error	20%

End Quiz Now

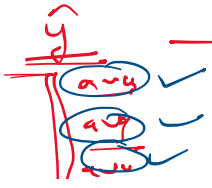
High error

Boosting works →

4 - trees

simplest possible model

#	Height	Gender	Weight(y)
1	1.6	M	82
2	1.5	F	55
3	1.4	F	61



2	1.5	F	55
3	1.4	F	61
4	1.4	M	65

← x → ← y →

y $\hat{y}$

pred^n $= \hat{y} = \bar{y}$

⇒ underfit

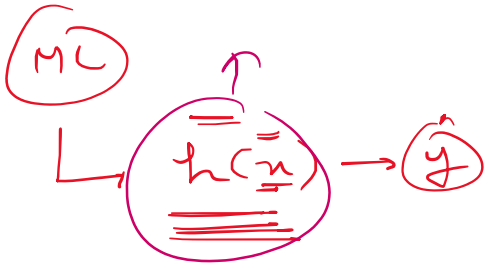
Step 0 → M_0 → High Bias + low variance

↳ simplest model

$h_0(x)$ ✓

$\{x_i \; y^i\}_{i=1}^{N}$ → fit model → mean model → M_0 (mean model)

$\hat{y} = \bar{y}$ (Pred^n always same)



(*) $h_0(x)$ → very Bad model

large errors

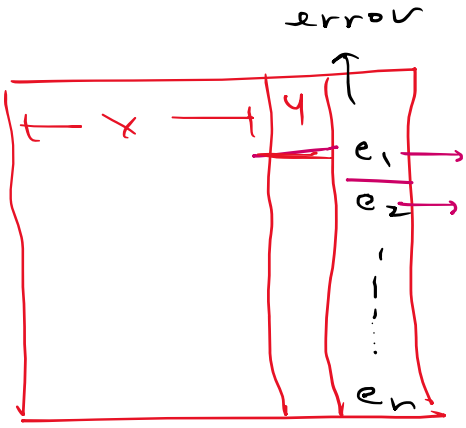
* for every D_i ⇒

① $err^i = y^i - \hat{y}^i$ → $\bar{y}$

$\bar{y}$ =

$h_0(x_i) = \bar{y}$

mean model



$h_0(x) = \hat{y}_i$ ⇒

$y^i = \hat{y}_i + err^i$

mean model

fit another model to predict these errors

②

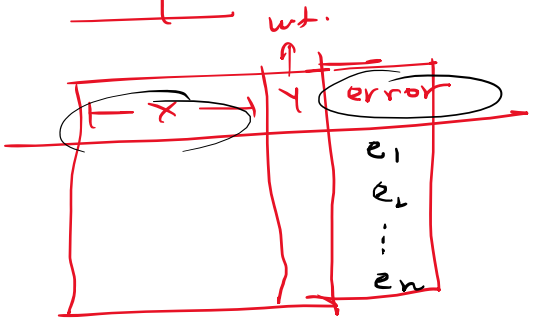
⇒ 0th model → lot of errors →

① M_1 ⇒ × not predicting well

what error M_0 has $h_0(x)$

↳ what error M_0 has made

Step 1 ⇒



M_0 → already trained

↳ Base / avg pred^n

↓

not good

M_1 → predict errors using i/p data

DTree with a low depth

$h_{m_1}(x)$ → $\{x^i, \text{error}^i\}_{i=1}^{N}$

(updated model)

actual pred'

$\Rightarrow y' - h_0(x')$

mean → $h_0(x)$

Baseline

(*) 2 models ⇒ $h_0(x)$ & $h_{m_1}(x)$

↓

telling what errors has made

ND ⇒

(*) DTree Regressor

M_1 → low depth → underfit model

↳ very simple DTree

final predn → weighted

(*) $\hat{y} = h_0(x) + \alpha_1 h_1(x)$

↳ predn errors

final model ⇒ end step

(*) $F_1(x) = h_0(x) + \alpha_1 \cdot h_1(x)$

↳

is F_1 a good model?

ND → large errors

Still underfit

1 more DTree → will try ∞

predict errors of combined model F_1

Step 0 → mean model → lot errors

1 → $h_1(x)$ → pred^n errors made By prev model

Combined model ⇒ $\hat{y}$ ⇒

latest model

$\bar{y}$ + error

Step − 2

combined

(*) $e^i = y^i - F_1(x^i)$

y^i → wt

F_1 → pred

$\{x^i, \text{error}^i\}$ → M_2 → $h_2(x)$

Bias ↑ (underfit)

errors made By previous models

(*) $F_1 \rightarrow h_0 + \alpha h_1$

M_2 → NO $F_1(x)$

$$F_2(x) = F_1(x) + \alpha_2 \cdot h_2(x)$$

* K = 100 ✓ — 1000 → overfit

* Reducing Bias

* low var.

* learnt during training ⇒

$$F_2(x) = h_0(x) + \alpha_1 h_1(x) + \alpha_2 h_2(x)$$

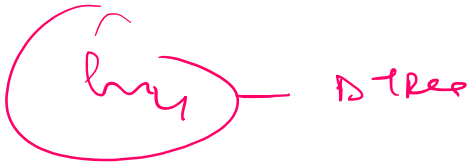
* Parameters

M_2 → underfitting

1 more tree → F_2 → errors

F_3 model

F_3 → underfitting



$\Rightarrow$ large training error will keep on $\downarrow$

$\Rightarrow$ k $\rightarrow$ no. of models

k steps

$$F_k(x) = h_0(x) + \alpha_1 h_1(x) + \alpha_2 h_2(x) + \cdots + \alpha_k h_k(x)$$

k $\rightarrow$ k = hyperparam

$$F_k(x) = h_0(x) + \sum_{i=1}^{k} \alpha_i h_i(x)$$

Quiz time!

Time Left: 14s

Why are we using DT with low depth here ?

18 users have participated

A	we are looking for high variance low bias mode	33%
✓ B	we are looking for high bias low variance model	56%
C	we are looking for overfit model	11%

5 min $\rightarrow$ 10:16

DOUBTS $\Rightarrow$

weight

$\Rightarrow$

B	we are looking for high bias low variance model	56%
C	we are looking for overfit model	11%
D	no reason. just picked it randomly.	0%

End Quiz Now

mean = =

Var — 100

height

Entropy

Random impurity

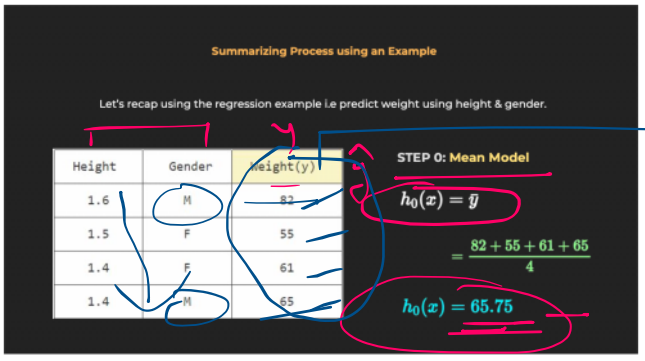
65.75 +16.25

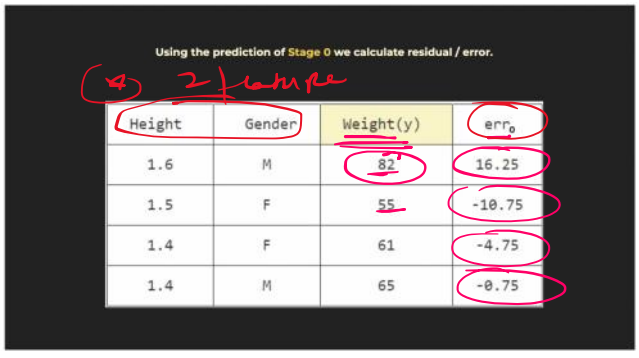
Var2

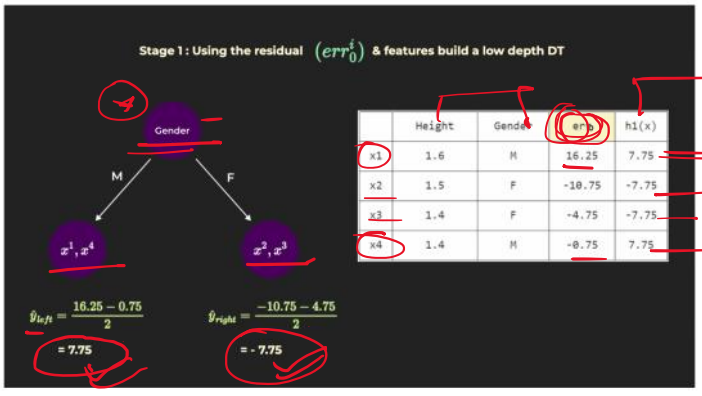
Depth = 1

gender — 2(q)

underfitted Tree



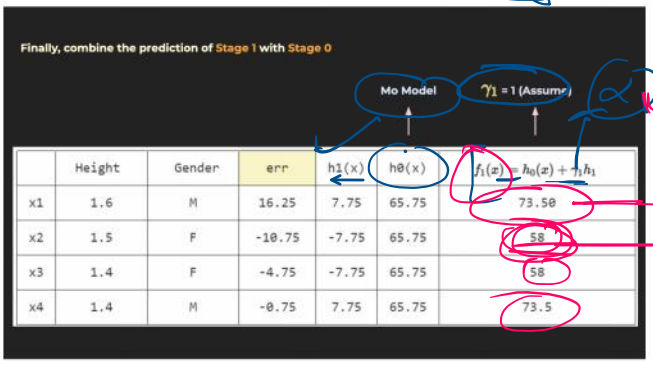




good errors

$h_0(x)$, $h_1(x)$

Stage 1



Right direction

error will be reduced

$H_2 \rightarrow F_2 \rightarrow F_1 + h_2$

2 (n) iterations

→ expect to see Bowden Dip

w_1, w_0

Hyperparam

Rate

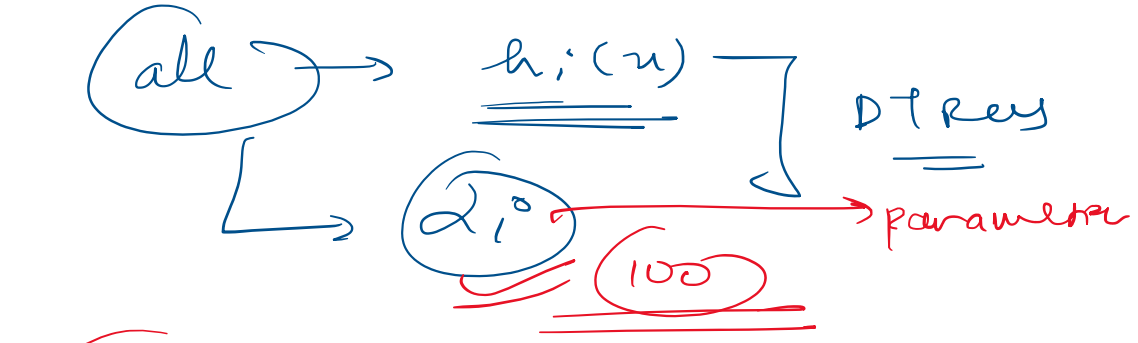
→ learnt during training

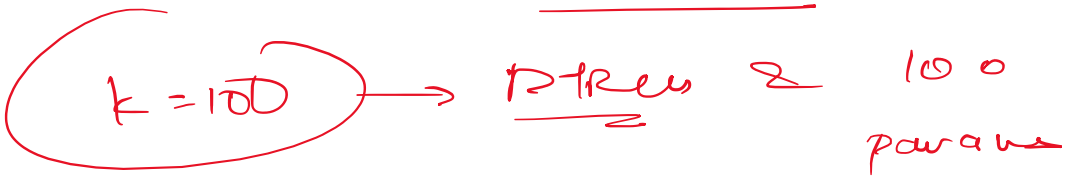
Hyperparam Tuning

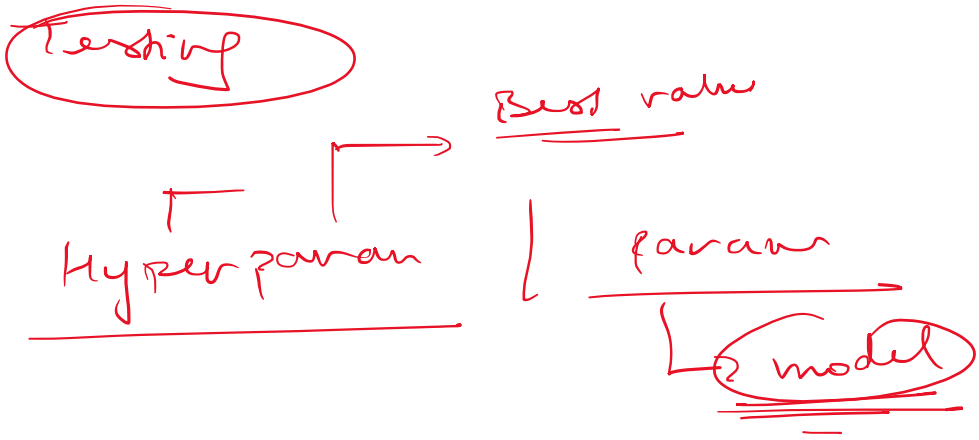
Boosting → gBoosting

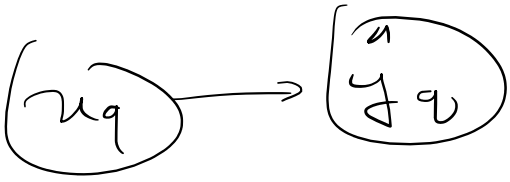
AdaBoost ✗

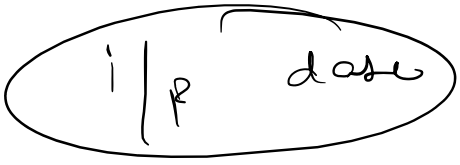
Train Boosting model





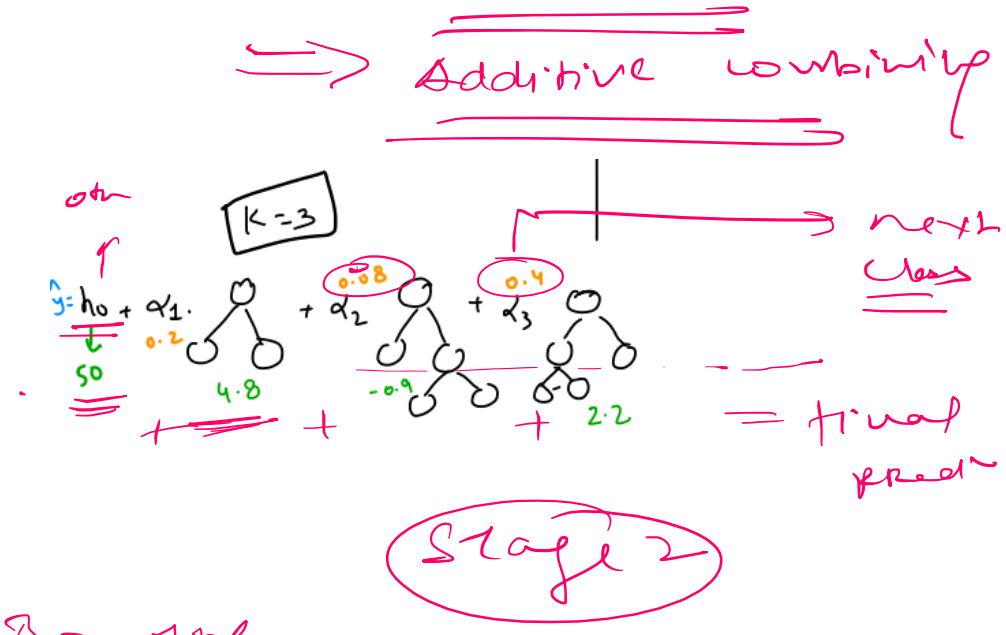


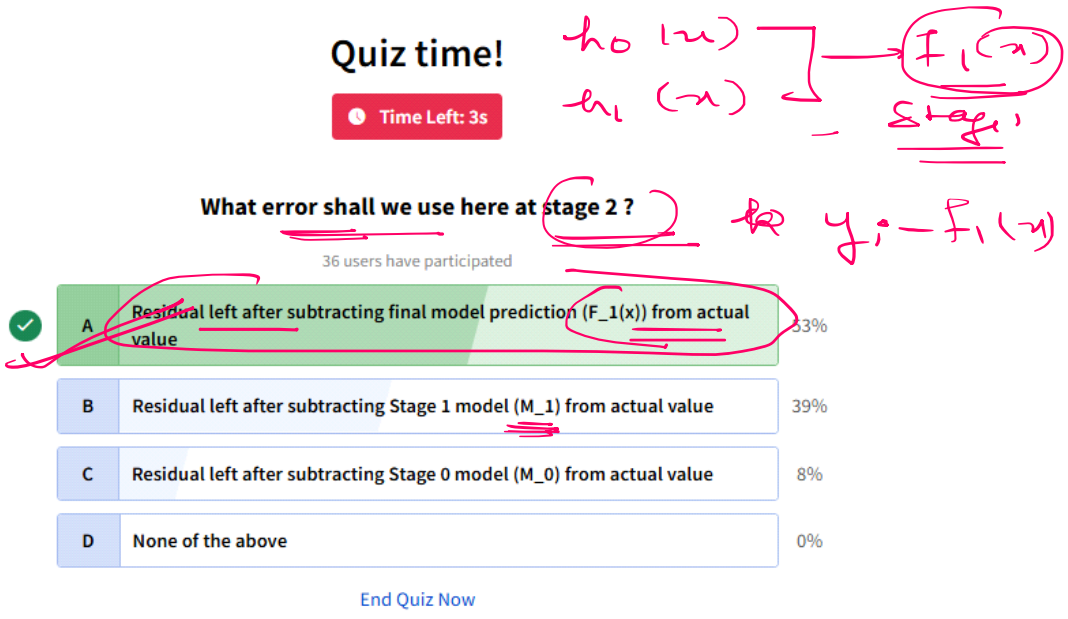




on model

$$F_k(x_q) = h_0(x_q) + \alpha_1 h_1(x_q) + \alpha_2 h_2(x_q) + \cdots + \alpha_k h_2(x_q)$$





$h_0(x)$, $h_1(x)$ → $F_1(x)$ — Stage 1

$y_i - F_1(x)$

h_0 → mean, $h_1(x)$ → errors

$F_1(x) = h_0 + h_1$

Pred wt Stage 1

Stage 2 $h = y_i - F_1(x)$

* PSEUDO RESIDUALS $\Rightarrow$ error $= y_i - \hat{y}_i$

Boosting $\rightarrow$ Residual concept

$$\text{Loss}(y, \hat{y}) = (y - \hat{y})^2$$

Sq. error

$\times N$

$\hat{y} = F_k(x)$

$\hookrightarrow$ final model at end of stage k

* $\frac{\partial L}{\partial} = \frac{\partial}{\partial} (y - \hat{y})^2$

$$\frac{\partial L}{\partial \hat{y}} = \frac{\partial}{\partial \hat{y}}(y - \hat{y})$$

$$= 2(y - \hat{y})(-1)$$

$$\frac{\partial L}{\partial \hat{y}} = -2(y - \hat{y})$$

-ve on Both sides

$$\Rightarrow -\frac{\partial L}{\partial \hat{y}} = 2(y - \hat{y})$$

error

Residual

$$-\frac{\partial L}{\partial F_k(x)} = 2(y - \hat{y})$$

-ve gradients

$\equiv$ Pseudo Residual

M_4

$M_4 \mid h_4(x) \Rightarrow \{x_i, err_i\}$

$M_4 \mid h_4(x) \Rightarrow \{x_i, err_i\}$

3 models already Trained

$F_3(x) = h_0 + \alpha_1 h_1 + \alpha_2 h_2 + \alpha_3 h_3$

h_4 x_i error

actual − F_3

$y_i - F_3(x_i)$

$y - \hat{y}$ → Regression

Pseudo Residual

Sq. loss

$-\frac{dL}{dF(x)} = 2(y - \hat{y})$

general term

(loss)

Gradient Boosting

log loss

$(y_i - \hat{y}_i) \Rightarrow$ Regressio

$(y_i - \hat{y}_i) \Rightarrow$ Regressio

loss func^n → RF ✗

(8) GBDT ⇒ loss → optimise

↳ practical ⇒

$M_j / h_j(x)$ → {x^i, error}

(j−1) models trained

✗ (actual − F_{j-1})

Sq. error

Classification

$\frac{-\partial L}{\partial F_{j-1}(x)}$ = pseudo Residual

more generic

Bagging → ✗

Quiz time!

Time Left: 7s

What do you think are the hyperparameter for boosting ?

35 users have participated

A	Number of trees	11%
B	Depth	9%
C	which loss to use	6%
D	All of the above	74%

End Quiz Now

